

Background

- **Drosophila melanogaster** is widely used in biological research because of its short life cycle and ~70% genetic similarity to humans.



Figure 1. Wild Type *D. melanogaster*

- **Microplastics (MPs)** are small plastic particles commonly found in **air, water, and food**, leading to frequent exposure in living organisms.



Figure 2. Microplastics

- MPs can disrupt **normal cellular function** and may increase **stress responses, cellular damage, and behavioral changes**.
- Previous research has shown effects such as **reduced movement** and altered sensory responses, suggesting impacts on **nervous system function**.
- Since memory depends on proper **neural signaling**, MP exposure may reduce **learning ability** and **memory retention**.
- However, the relationship between **MP concentration** and **short-term memory** is still not well understood.
- This study investigates how different levels of microplastic exposure affect short-term memory in **Drosophila melanogaster**.

Objectives

Research Question: How do varying concentrations of PET microplastics affect short-term memory in *Drosophila melanogaster*?

Hypothesis: If *D. melanogaster* are exposed to increasing concentrations of microplastics, then memory performance will decrease.

Variables:

- **Independent Variable:** Microplastic concentrations (1 mg/L, 5 mg/L, 10 mg/L)
- **Dependent Variable:** Percentage of correct maze choices, and the percentage of maze completion
- **Control:** *Drosophila melanogaster* raised and tested under identical conditions without microplastic exposure.
- **Constants:** Fly strain, feeding conditions, maze setup, odor concentrations, and test duration (2 minutes).

Effects of Oral Exposure to PET Microplastics on Short-Term Memory in *Drosophila melanogaster*

Methodology

Making Microplastics and Y-Maze Assays

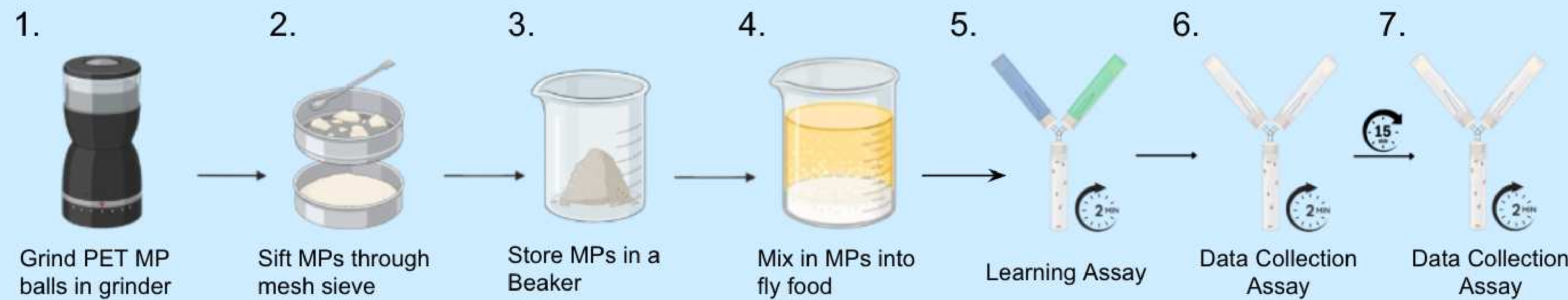


Figure 3. Illustration of Experimental Design (Made in BioRender)

Collecting Data

8. Condition flies to associate a specific odor with the correct path in the Y-maze using odor and color cues.
9. Immediately transfer flies into the data collection Y-maze following conditioning.
10. Allow flies to explore the maze freely for 2 minutes.
11. Record the number of flies that:
 - a. Choose the correct conditioned arm
 - b. Successfully complete the maze by entering either choice arm
12. Count the number of flies present in each odor vial after the trial is completed.
13. Compile all trial data into a spreadsheet and calculate:
 - a. Percentage of correct choices
 - b. Percentage of maze completion
14. Perform statistical analysis, (ANOVA test) to compare memory performance across microplastic concentrations.



Figure 4. Grinding of MPs



Figure 5. Data Collection Assay

Results

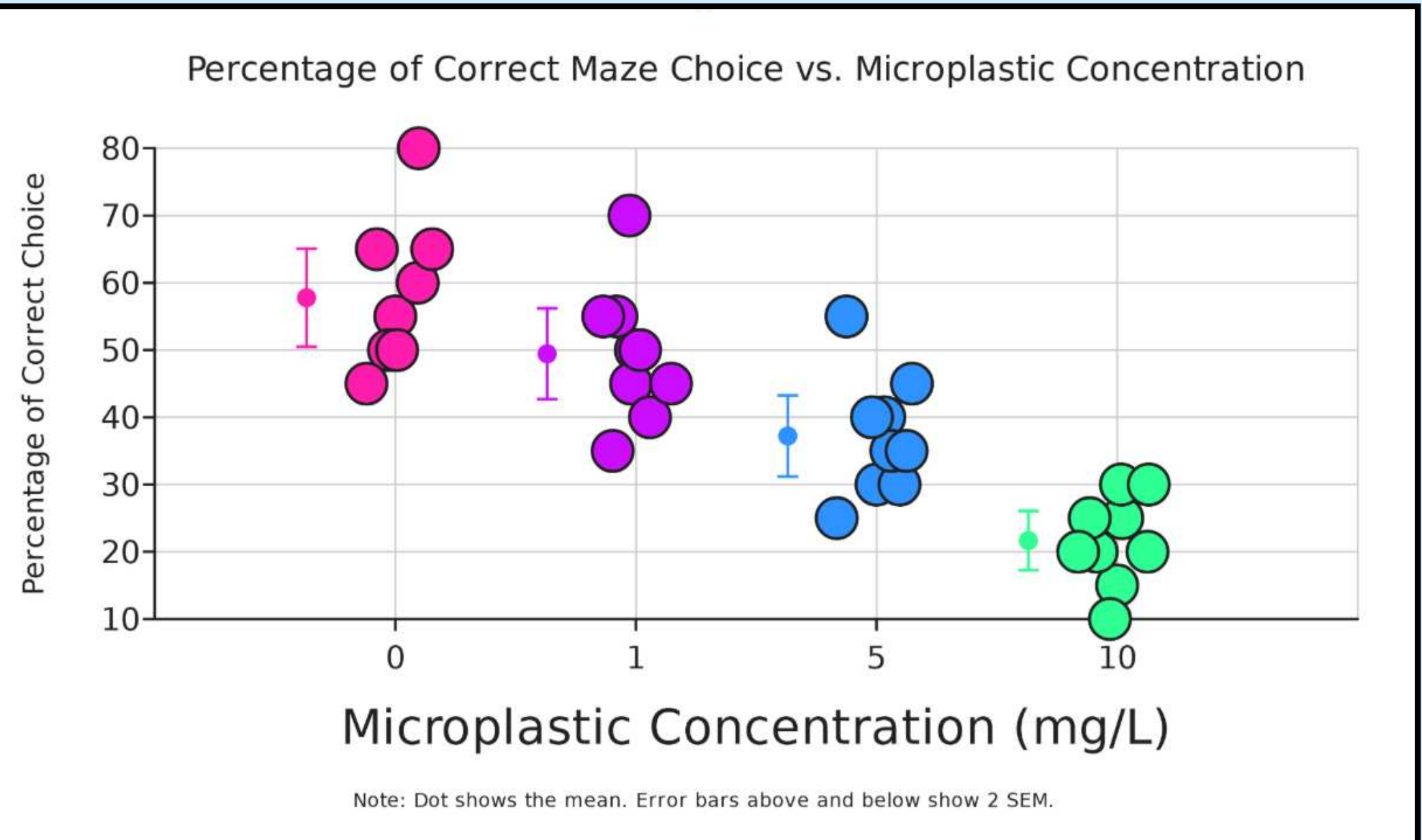


Figure 6. Memory Function Results for *Drosophila melanogaster* (Data Classroom)

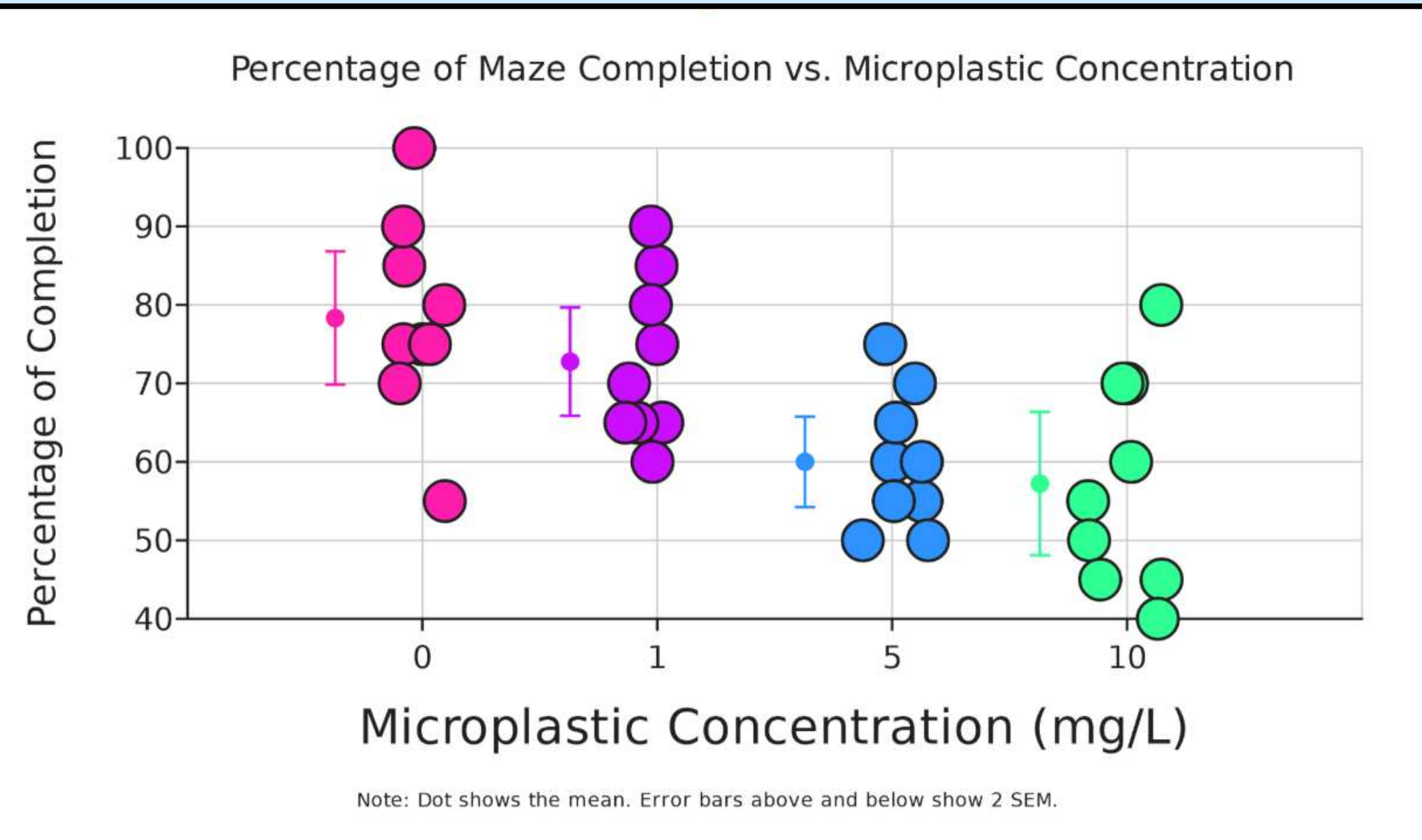


Figure 7. Motor Function Results for *Drosophila melanogaster* (Data Classroom)

- 0 mg/L of MPs
- 1 mg/L of MPs
- 5 mg/L of MPs
- 10 mg/L of MPs

Key Takeaway:

As the microplastic concentration in fly food increased, the percentage of flies that were able to correctly choose the right side and complete the maze decreased.

Stastics:

Results were subject to an ANOVA test.

Memory Performance

- Microplastic concentration showed a **statistically significant difference** in memory performance, **p < 0.01**
- ANOVA results: F(3,32) = 26

Motor Function

- Microplastic concentration showed a **statistically significant difference** in maze completion rate, **p < 0.01**
- ANOVA results: F(3,32) = 6.9

Conclusion:

- The **hypothesis was supported** as increasing microplastic concentration led to decreased memory performance, showing a clear **dose-dependent relationship**.
- Flies exposed to the **highest concentration (10 mg/L)** showed the lowest percentage of correct maze choices compared to the control group.
- Statistical analysis confirmed that microplastic concentration had a **significant effect** on short-term memory in *Drosophila melanogaster*.
- **Motor function was also negatively affected** at higher microplastic concentrations.
- These findings suggest that microplastics may negatively affect **neurological function and behavior**.
- Future studies should include a larger sample size and additional trials to improve reliability.
- Further research could investigate:
 - Effects on **long-term memory**
 - Different types of microplastics
 - Neurological mechanisms involved in memory impairment

References

All images are not cited are take/created by the student or from Canva.

Davis, R. L. (2023). Learning and memory using *Drosophila melanogaster*: A focus on advances made in the fifth decade of research. *Genetics*, 224(4). <https://doi.org/10.1093/genetics/ygad085>

Inami, S., Sato, S., Kondo, S., Tanimoto, H., Kitamoto, T., & Sakai, T. (2020). Environmental light is required for maintenance of long-term memory in *Drosophila*. *The Journal of Neuroscience*, 40(7), 1427-1439. <https://doi.org/10.1523/JNEUROSCI.1282-19.2019>

Jayavel, S., Govindaraju, B., Michael, J. R., & Viswanathan, B. (2024). Impacts of micro and nanoplastics on human health. *Bulletin of the National Research Centre*, 48(1), 1-19. <https://doi.org/10.1186/s42269-024-01268-1>

Kruger, L., & Denton, T. (2020). A standardized method for incorporation of drugs into food for use with *Drosophila melanogaster*. *Analytical Biochemistry*, 599, 113740. <https://doi.org/10.1016/j.ab.2020.113740>

Mohandas, R., Thakare, M., Sunke, S., Iqbal, F. M., Sridharan, M., & Das, G. (2022). Enhanced olfactory memory detection in trap-design Y-mazes allows the study of imperceptible memory traces in *Drosophila*. *Learning & Memory*, 29(10), 355-366. <https://doi.org/10.1101/m.053545.121>

Schulz, J., Franz, H. R., Deimel, S. H., & Widmann, A. (2024). Exploring neonicotinoid effects on *Drosophila*: Insights into olfactory memory, neurotransmission, and synaptic connectivity. *Frontiers in Physiology*, 15, 1363943. <https://doi.org/10.3389/fphys.2024.1363943>

Silva, V., Palacios-Munoz, A., Okey, Z., Adair, K. L., Waddell, S., Douglas, A. E., & Ewer, J. (2021). The impact of the gut microbiome on memory and sleep in *Drosophila*. *The Journal of Experimental Biology*, 224(5), jeb233619. <https://doi.org/10.1242/jeb.233619>

Tolwinski, N. S. (2024). *Drosophila*—A model system for developmental biology. *Journal of Developmental Biology*, 12(2), Article 15. <https://doi.org/10.3390/jdb12020015>

Wang, Y., & Jiang, Y. (2024). *Drosophila melanogaster* as a tractable eco-environmental model to unravel the toxicity of micro- and nanoplastics. *Environment International*, 192, 109012-109012. <https://doi.org/10.1016/j.envint.2024.109012>

Zamberlan, D. C., Halmenschelager, P. T., Silva, L. F. O., & da Rocha, J. B. T. (2020). Copper decreases associative learning and memory in *Drosophila melanogaster*. *Science of the Total Environment*, 710, 135306. <https://doi.org/10.1016/j.scitotenv.2019.135306>